

Predicting Diabetes Risk from Behavioral and Socioeconomic Indicators

Group 33



Section A

Motivation, Question, and Data



The Problem

Why does early diabetes detection matter?

- 37M+ Americans have diabetes
- 96M adults have prediabetes
- Costs \$327B annually in economic burden
- Type 2 diabetes is largely preventable

Yet, many cases still go undetected as clinical testing is expensive and inconsistently applied.

A low-cost, survey-based model could fill this gap



Research Question:

To what extent can diabetes status be predicted from observable health behaviors and socioeconomic characteristics — and which factors are most strongly associated with increased risk?



The Data

CDC Behavioral Risk Factor Surveillance System (BRFSS) 2015

253,680

survey respondents

21

predictor variables

0

missing values

13.9%

diabetes prevalence

Clinical & Physiological

- BMI
- High Blood Pressure (HighBP)
- High Cholesterol (HighChol)
- Stroke history
- Heart disease/attack

Behavioral & Lifestyle

- Physical activity (past 30 days)
- Fruit & vegetable consumption
- Smoking history
- Heavy alcohol use

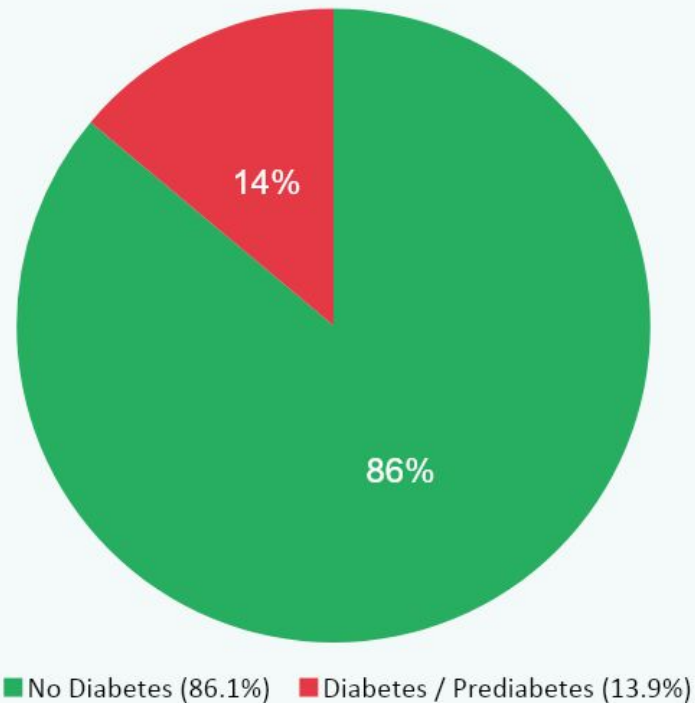
Self-Reported Health

- General health rating (GenHlth, 1–5)
- Poor mental health days
- Poor physical health days
- Difficulty walking (DiffWalk)

Demographics & SES

- Age (13 categorical bins)
- Sex, Education
- Income (8-level ordinal)
- Healthcare access & cost barriers

Target Variable & Class Imbalance



Why this matters for modeling

Accuracy is misleading

A classifier that predicts 'no diabetes' every time achieves 86% accuracy — yet catches zero cases.

Sensitivity matters most

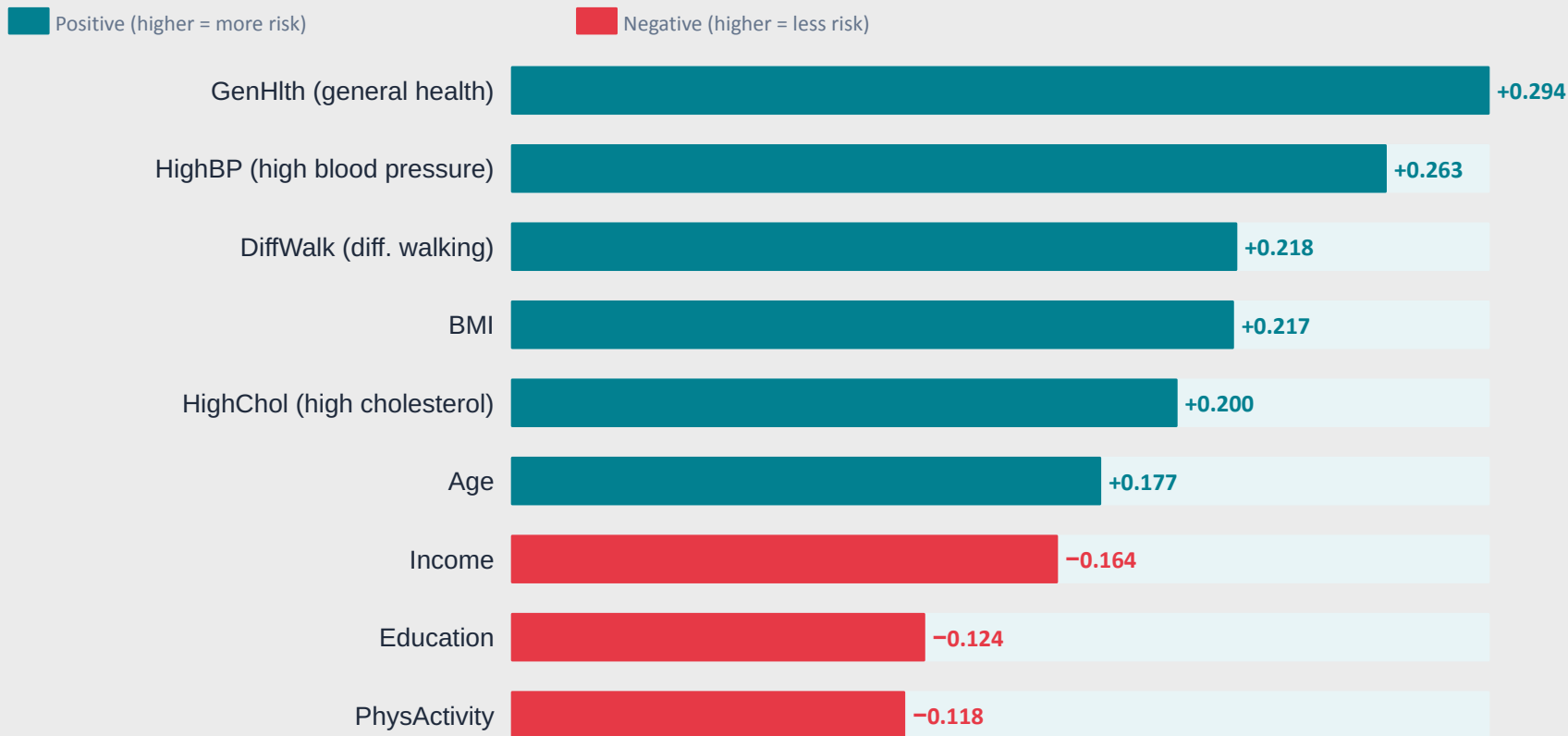
In a screening context, missing a true diabetic (false negative) is far more costly than a false alarm.

We prioritize AUC + cost-weighted thresholds

AUC is threshold-independent and robust to imbalance. We also test a 6:1 cost ratio (FN:FP).

What Does the Data Tell Us? (EDA Preview)

Pearson correlations with diabetes status — ranked by absolute magnitude



Section B

Methods and Analytic Strategy



Step 1: Data Preparation

Train / Test Split

Training 70%
177,576 obs

Test 30%
76,104

set.seed(471)

Fixed seed ensures full reproducibility

All model fitting, LASSO tuning, and hyperparameter selection performed on training set only.

Final metrics reported on held-out test set which were never seen during training.

Class Imbalance Problem

No Diabetes 86.1%

13.9%

The trap: predicting 'No Diabetes' for everyone = 86% accuracy but 0% recall.

Our solution involves two approaches:

A Standard models (baseline)

B Class-reweighted models
(diabetic cases weighted ~6x)

Both evaluated on same test set for fair comparison.

Step 2: Variable Selection via LASSO

21 candidate predictors → 13 selected

Why LASSO?

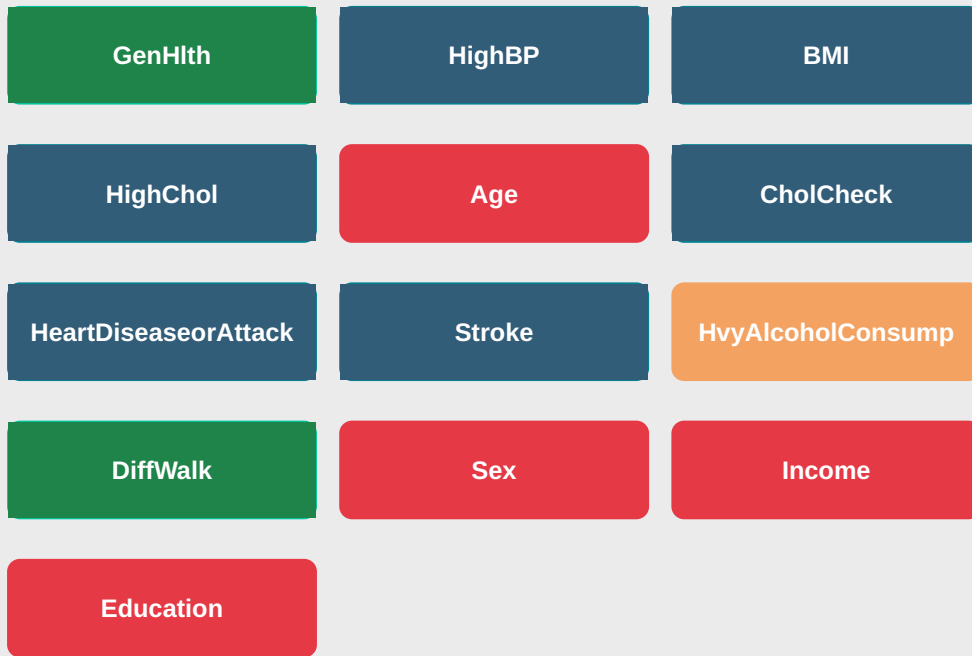
L1 penalty shrinks weak predictors exactly to zero
— automatic variable selection.

Many predictors are correlated (*HighBP*, *HighChol*, *HeartDisease* all cluster together)
— LASSO picks one representative.

λ selected via 10-fold cross-validation.

We use **lambda.1se**, the largest λ within 1 SE of minimum error, for a sparser, more conservative model suited for screening.

13 Variables Selected



Clinical

Self-reported health

Behavior

Demographic/SES

Step 3: Six Models, Two Approaches

Each model run twice — once standard, once with class reweighting

Logistic Regression

Interpretable baseline. Coefficients = odds ratios. Clinically communicable.

LASSO Logistic

Variable selection + inference combined. Identifies the 13 key predictors.

Classification Tree

Reveals threshold effects & interactions. Visual decision rule.

Random Forest

500 trees voted. Reduces variance. Variable importance ranking.

XGBoost

Sequential error correction. Best AUC. Industry standard for tabular data.

Neural Network

Captures complex nonlinear patterns. Competitive AUC on reweighted data.

 Interpretable

 Ensemble

 Flexible

Step 4: How We Evaluate the Models

Confusion Matrix

	Pred: No	Pred: Yes
Act: No	TN	FP
Act: Yes	FN (worst!)	TP

*FN = False Negative: we said 'healthy' but they have diabetes.
This is the error we most want to minimize.*

Our Four Key Metrics

AUC

Area Under ROC Curve

Primary ranking metric. Threshold-independent and robust to imbalance. Measures how well the model separates diabetics from non-diabetics across ALL thresholds.

Recall (Sensitivity)

$TP / (TP + FN)$

Most critical for screening. Of all true diabetics, what % did we catch? Target: ~80% in reweighted models.

Precision

$TP / (TP + FP)$

Of those flagged, what % truly have diabetes? Lower in reweighted models but acceptable tradeoff for screening.

Accuracy

$(TP + TN) / n$

Least useful here due to imbalance. A model always predicting 'No' gets 86% so we deprioritize this metric.

Section C

Findings, Interpretation, and Conclusion



Evaluation Metrics of Weighted / Unweighted

Models	Recall	ROC/AUC	Accuracy	Precision
Logistic Regression	0.761 / 0.157	0.825 / 0.823	0.732 / 0.866	0.313 / 0.551
Decision Tree	0.780 / 0.109	0.772 / 0.723	0.696 / 0.866	0.286 / 0.593
Random Forest	0.819 / 0.498	0.759 / 0.942	0.620 / 0.926	0.245 / 0.936
XGBoost	0.791 / 0.066	0.828 / 0.827	0.720 / 0.866	0.306 / 0.673
Neural Network	0.800 / -	0.828 / -	0.718 / -	0.306 / -



Conclusions

Limitations

Self-reported survey bias

Recall errors, social desirability bias

Loss of detail

Binary medical indicators obscure severity

Cross-sectional data

Cannot infer causality, only association

Class imbalance trade-off

Tradeoff of recall and precision

Model performance ceiling

Weighted models converged around 0.82 AUC

Final Remarks

Survey-based ML can be a **powerful tool** for early detection and public health targeting.

GenHlth, HighBP, BMI, Age, HighChol proved to be **key predictors** across all models.

Random Forest is useful for initial screening; **Neural Network** is recommended for deeper analysis

Future Work could include using more detailed clinical features such as continuous blood pressure or glucose levels.

Thanks!

Any Questions?